

SANTOSH PARMAR

Embedded Software Engineer | Embedded Linux | C/C++ | Telematics | GNSS/GSM | CAN | MQTT | AWS

PROFESSIONAL SUMMARY

Embedded Software Engineer with 4+ years of hands-on experience in Embedded C/C++, Embedded Linux, telematics, device diagnostics, GNSS/GSM integration, remote updates, cloud-connected telemetry, and field testing. Experienced in designing and developing production-oriented applications for hardware data collection, telemetry logging, modem/GNSS communication, remote diagnostics, JSON-based data pipelines, MQTT communication, and AWS integration. Strong foundation in Linux system programming, multithreading, debugging, shell scripting, and embedded communication protocols.

CORE SKILLS

- Languages: C, C++, Python, Shell Scripting
- Embedded Systems: Embedded Linux, STM32, RTOS, UART, I2C, SPI, GPIO, PWM, ADC
- Protocols & Connectivity: MQTT, TCP/UDP Socket Programming, GSM, GNSS, CAN
- Linux & Debugging: Linux System Programming, GDB, Valgrind, Bash, Cross Compilation
- Build Tools: Git, CMake, Makefile, CI/CD
- Cloud / Storage / Data: JSON, AWS SDK (S3, Lambda), SQLite
- ATS Keywords: Embedded C, C++, Firmware Development, Telematics, Multithreaded Application Development, Diagnostics, Remote Update, OTA Update, Device Monitoring, Sensor Data Logging

PROFESSIONAL EXPERIENCE

Embedded Software Engineer | ZF Group | Bangalore, India | Oct 2023 – Present

- Developed telematics applications in C/C++ for embedded Linux systems to collect low-level hardware and device data including GSM, GNSS, power, RAM, CPU, eMMC, temperature, and motion sensor information.
- Built and enhanced remote diagnostic features to support field issue analysis, health monitoring, and device troubleshooting.
- Developed a PC-based diagnostic application to communicate with field devices remotely for analysis and service support.
- Worked on telematics logging systems and maintained robust communication with cloud services using MQTT and AWS SDK.
- Contributed to telemetry acquisition, structured data generation, logging pipelines, and software reliability improvements for connected embedded devices.

Firmware Developer (Project-Based Work) | ZF Group | Jun 2022 – Oct 2023

- Developed geo-fencing functionality for an existing legacy product to support location-aware behavior and customer-specific features.
- Implemented dual-profile logic in a legacy product to support new hardware design changes and help reduce R&D; cost.
- Contributed to multiple POCs and field tests across embedded Linux platforms.
- Delivered customer-oriented feature development and product changes based on project requirements.

KEY PROJECTS

AWS-Connected Telematics Logger | Oct 2024 – Present

- Designed and developed a multithreaded C++ application to log GSM, GPS, accelerometer, gyroscope, and power data from an embedded Linux board.
- Implemented S3 upload through MQTT and direct AWS SDK integration, including file chunking and retry handling for reliable telemetry upload.
- Built a circular buffer-based file management system with 30-day retention policy and 32 MB file-size cap to support long-running deployments.

- Tech Stack: C++, Linux, Shell, AWS (S3, Lambda), MQTT, JSON

GSM + GNSS Logger with Fault Simulation UI | Jun 2024 – Oct 2024

- Built a Qt5-based diagnostic UI to display real-time GNSS and GSM data in table views for validation and debugging.
- Added fault simulation logic to induce segmentation faults and validate recovery behavior.
- Improved visibility into modem/GNSS behavior and fault-handling scenarios through real-time diagnostics.

Remote Application and BSP Firmware Update | Jan 2024 – Jun 2024

- Contributed to a remote update mechanism for application and BSP firmware to deploy new features in field devices.
- Reduced manual effort and field dependency by enabling remote update workflows.

Field Test Automation | Feb 2023 – Jan 2024

- Designed automated test setups for evaluating embedded devices under varied power conditions and GPS/GSM performance scenarios.
- Created structured field reports and comparison-based analysis to improve validation quality.
- Used automation for repeatable field test workflows and log-based analysis.

Power & Boot Status Monitoring System | 2023

- Integrated board power metrics, uptime, and boot reasons into structured JSON objects with persistent logs for analysis.
- Improved observability of system health and boot behavior on embedded devices.

EDUCATION

- B.Tech – Electrical & Electronics Engineering | HKBK College of Engineering | Jul 2017 – Aug 2021

CERTIFICATIONS

- Embedded System, C/C++, Linux – Vector India
- C/C++ Programming – NPTEL
- Linux Shell Scripting – Udemy